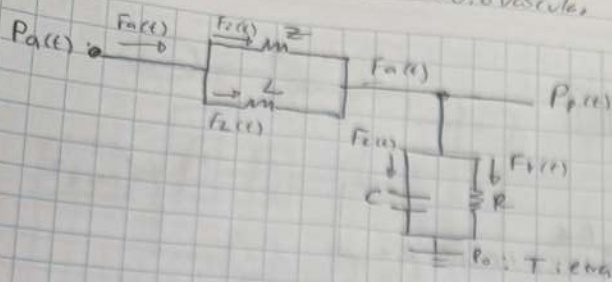


Practica 5.4 : Sistema Cardiovascular

10-10-2015



Ecuación principal

$$F_1(t) = F_2(t) + F_3(t) \equiv F_c(t) + F_r(t)$$

$$F_2(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_3(t) = C \frac{dP_p(t)}{dt}$$

$$F_1(t) = \frac{1}{L} \int [P_a(t) - P_p(t)] dt$$

$$F_r(t) = \frac{P_p(t)}{R}$$

Procedimiento algebraico

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt = C \frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{Ls} \right) P_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls} \right) P_p(s) = \left(\frac{CSR + 1}{R} + \frac{Ls + Z}{ZLs} \right) P_p(s)$$

$$\left(\frac{Ls + Z}{LZs} \right) P_a(s) = \left(\frac{CRLs^2 + LZs + RLS + RZ}{RZLs} \right) P_p(s) = \left(\frac{CRLZs^2 + LZs + LSR + RZ}{RZLs} \right) P_p(s)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{Ls + Z}{LZs}$$

$$CRLs^2 + LZs + RLS + RZ$$

$$RZLs$$

$$\frac{P_p(s)}{P_a(s)} = \frac{Ls + Z (R)}{CRLs^2 + LZs + RLS + RZ (1)}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{LSR + RZ}{CRLs^2 + LZs + RLS + RZ}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{(Ls + Z) R}{CRLs^2 + (LZ + RLS) + RZ}$$

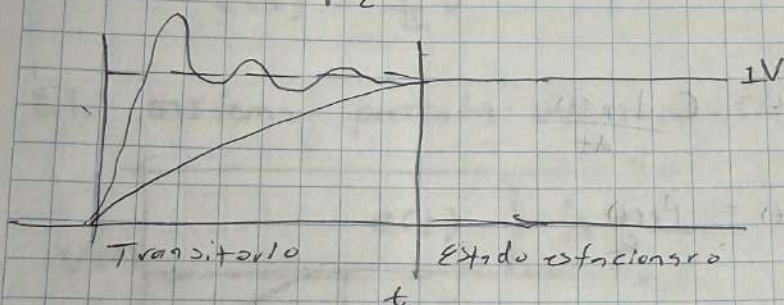
$$\frac{P_p(s)}{P_a(s)} = \frac{Ls + Z (RZLs)}{CRLs^2 + LZs + RLS + RZ (LZs)}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s P_a(s) \left[1 - \frac{P_e(s)}{P_a(s)} \right]$$

$$= \lim_{s \rightarrow 0} s \times \frac{1}{s} \left[1 - \frac{R_L s^2 + R_Z}{CLR_Z s^2 + (L_Z + RL)s + R_Z} \right]$$

$$= 1 - \frac{R_Z}{R_Z} = 0V$$



Estabilidad en Lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = CLR_Z$$

$$b = L_Z + RL$$

$$c = R_Z$$

$$\lambda_{1,2} = \frac{-(L_Z + RL) \pm \sqrt{(L_Z + RL)^2 - 4(CL R_Z^2)}}{2 CL R_Z}$$

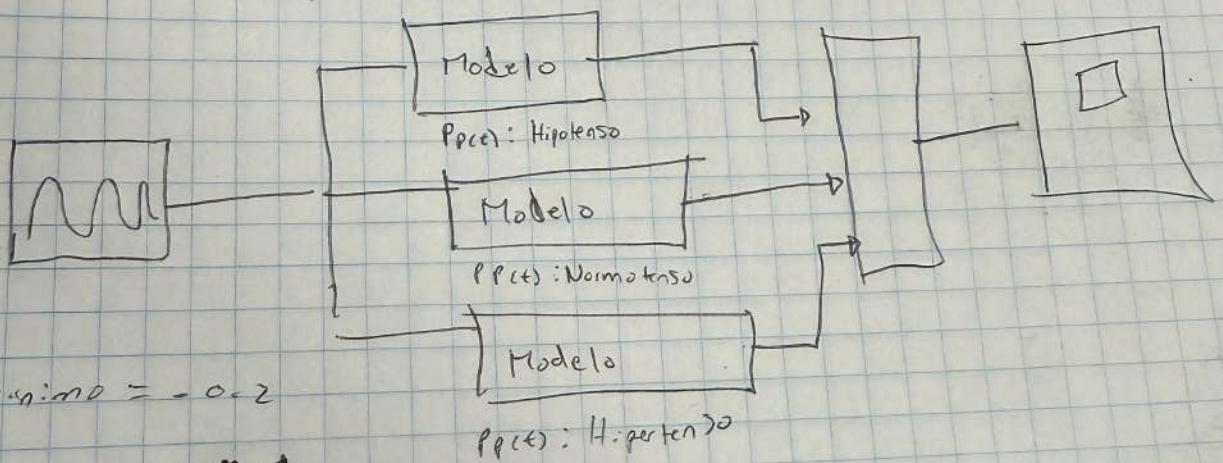
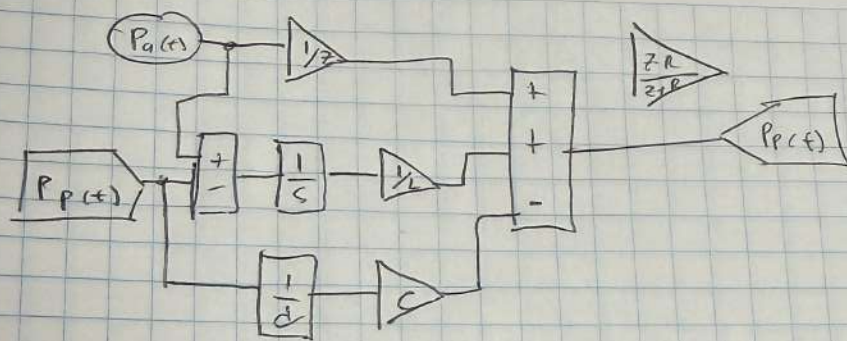
El sistema tiene una respuesta estable

Porque $\text{Re } \lambda_{1,2} < 0$

Modelo de ecuaciones integro-diferenciales

$$P_p(s) \left(\frac{1}{R} + \frac{1}{Z} \right) = \frac{P_a(t)}{Z} + \frac{1}{L} \int (P_a(s) - P_p(s)) dt - \frac{Cd P_p(s)}{dt}$$

$$P_p(s) = \left(\frac{P_a(s)}{Z} + \frac{1}{L} \int [P_a(s) - P_p(s)] dt - \frac{Cd P_p(s)}{dt} \right) \frac{ZR}{Z + R}$$



minimo = -0.2

maximo = ~~0.1~~ 1

Seto : 106